SECTION 1

Frame the problem and look at the big picture

- **Business objective:** Define success in business terms, how the solution will be used, and how it compares to current workarounds before choosing supervised, unsupervised, or online formulations.
- **Measurement:** Pick performance metrics aligned with the business goal, note minimum viable accuracy, and list assumptions to validate with experts or manual prototypes.
- **Reuse:** Identify comparable problems, available human expertise, and tools that can shorten the path.

SECTION 2

Get the data

- **Acquisition plan:** List required data volumes, document sources, storage needs, legal constraints, and access credentials in a dedicated workspace.
- **Ingestion hygiene:** Convert to manipulable formats without corrupting content, anonymize or remove sensitive fields, and characterize types (time series, geo, etc.).
- **Test isolation:** Draw a held-out test set early and never peek at it during exploration or tuning.

SECTION 3

Explore the data

- **Structured exploration:** Copy data for analysis (sample if huge), record work in a notebook, and profile each attribute's type, missingness, noise, and usefulness.
- **Insight building:** Identify targets for supervised tasks, visualize distributions and correlations, and note manual solutions that hint at viable features.
- **Documentation:** Capture promising transformations and gaps that send the project back to data collection.

SECTION 4

Prepare the data

- **Reproducibility:** Work on copies and encode every transformation in reusable functions for training, validation, test, and production paths.
- **Cleaning and features:** Handle outliers and missing values, drop useless columns, engineer aggregates and decompositions (dates, categoricals, nonlinear terms).
- **Scaling:** Standardize or normalize features so algorithms see comparable ranges.

SECTION 5

Shortlist promising models

- **Rapid benchmarks:** On large data, subsample for quick trials across linear, naive Bayes, SVM, forests, neural nets, etc., with default hyperparameters.
- **Rigorous comparison:** Use cross-validation means and spreads, study important variables and error types, and iterate brief feature-engineering rounds.
- **Selection:** Keep three to five diverse top models whose errors differ, improving robustness of the final choice.

SECTION 6

Fine-tune the system

- **Hyperparameter search:** Treat preprocessing choices as tunables; prefer random search or Bayesian optimization on ample training data with cross-validation.
- **Ensembles:** Combine strong models when they beat individuals alone.
- **Generalization check:** Measure once on the pristine test set and do not tweak afterward to avoid test-set overfitting.

SECTION 7

Present your solution

- **Communication:** Document the pipeline, lead with the big picture, and explain how results meet the business objective.
- **Honesty:** Report what worked and failed, state limitations, and support claims with clear visuals or memorable facts stakeholders can recall.

SECTION 8

Launch, monitor, and maintain

- **Production readiness:** Wire live inputs, unit tests, and deployment automation; monitor live performance with alerts for drops or data drift.
- **Operational risks:** Watch input quality (broken sensors, stale upstream feeds) especially in online learning; plan human-in-the-loop evaluation where labels lag.
- **Retraining:** Retrain on fresh data on a schedule, automating pipelines so models do not rot as the world changes.